Capstone Project Progress Reports

There are four progress reports and a final report for the iBioMed Capstone Project. The four progress reports are short, structured updates that show how your team's project is developing throughout the year. Each report should explain what has been completed, what has been learned, what challenges remain, and what the team plans to do next.

The reports are intended to provide meaningful checkpoints and feedback, not to evaluate the final quality of the design. They are graded on the quality of the progress update itself, including evidence of project development, clear communication, appropriate use of engineering methods, and a realistic plan for moving forward. The details of each of the report can be found below.

Pass/Fail and remediation

- The reports will be graded by teaching assistants (TAs) on a pass/fail basis. The grading of the reports is meant to provide feedback on current work and suggestions for future work in the project.
- A Pass indicates that the submission meets the expectations and learning outcomes described for this assessment.
- A Fail indicates that submission does not yet meet the standard and must be revised and resubmitted within one week after feedback is received.
- A Fail does not directly reduce the final course grade. Feedback and remediation are intended to improve the project and the Final Project Report.

Each progress report should include a title page with the following information:

- A one-sentence descriptive project title (not a one-word title or simply 'Capstone Project')
- Team number and, if used, team name
- Stakeholder(s), co-supervisor(s), and instructor(s)
- Each group member's first and last name, MAC ID, and program (e.g. ELECBME)

Formatting and submission checklist

- Follow the IBEHS Writing Guidelines posted on Avenue for formatting of reports.
- Use Times New Roman, a minimum 11-point font, single spacing, justified alignment, and standard 1-inch (2.5 cm) margins.
- Place page numbers in the footer and your team number in the right header.
- Number and reference every figure and table in the text. Use clear, readable figures and descriptive captions.
- Use a clear heading structure, academic language, and careful proofreading. Formatting penalties may be applied.
- Appendices may contain supporting information such as calculations, technical drawings, detailed schematics, supplementary data, code excerpts, or other evidence. Information essential to understanding or evaluating the submission must remain in the report body.
- Generative AI may not be used to produce final technical submissions or written deliverable content. Permitted preliminary uses, such as brainstorming and editing, must follow the course outline and assessment-specific guidance and must be cited if used.
- Submit the required file(s) through Avenue by the posted deadline. Avenue instructions and announcements take precedence if administrative details change.

Assessment of Progress Reports

- Each Progress Report will be assessed based on the extent to which the applicable submission requirements have been addressed completely, correctly, and with appropriate supporting evidence.
- If content in any section is vague, incomplete, or insufficiently supported, a minor revision and resubmission may be required.
- If required content is absent, unsupported, fundamentally incorrect, or demonstrates a significant misunderstanding, a major revision and resubmission will be required.
- Each report will also be assessed on the quality of its written and technical communication, including clarity, concision, organization, professional formatting, appropriate use of figures and evidence, and clear identification of relevant assumptions and limitations.

IBEHS 5P06: Progress Report 1 – Background, Motivation, and Needs

Due Date: Friday, October 2, 2026
Length: 1 – 2 pages, excluding the title page and references

Progress Report 1 establishes why your project matters, whom it serves, and how your team will recognize meaningful progress. It should demonstrate an evidence-based understanding of the stakeholder and user context while identifying early risks, uncertainties, and project barriers.

Submission requirements

1. **Background and motivation**
   - Describe the problem context and the circumstances that motivate the project.
   - Use stakeholder discussions and relevant literature to explain why the problem is important and why an engineering response is warranted.
   - Define the intended user(s), stakeholder(s), and setting of use with enough specificity to guide later design decisions.
2. **User needs and project statement**
   - Summarize the user needs that your team has identified and distinguish needs from proposed solutions.
   - Explain the expected impact of a successful design on the stakeholder, broader society, or environment. Consider health, safety, economic, social, environmental, legal, and cultural dimensions where relevant.
   - End with a short, explicit need statement that is visually distinct from the surrounding text (e.g. bolded, italicized, on a separate line etc.).

IBEHS 5P06: Progress Report 2 – Design Inputs

Due Date: Friday, October 23, 2026
Length: 2 – 3 pages, excluding the title page and references
Required consultation: Agree on the design-process structure with your supervisor(s) before submission

Progress Report 2 translates the user needs established in Progress Report 1 into design inputs and documents how your team will explore the solution space. If the user profile or need statement has changed, briefly explain the new information and update the framing before presenting the design inputs and design process**.**

Submission requirements

1. **Updated user and problem framing**
   - If applicable, briefly identify any significant changes to the user profile, stakeholder context, need statement, or project scope since Progress Report 1.
   - Explain the evidence or feedback that led to each change. If nothing significant has changed, state that the original information from Progress Report 1 remains current and restate the problem statement.
2. **Design inputs**
   - Define the requirements, specifications, objectives, and constraints that the design is expected to satisfy.
   - Develop objective and, wherever feasible, quantitative criteria through stakeholder consultation, literature review, and discussions with disciplinary instructors.
   - Consider technical, economic, environmental, ergonomic, societal, safety, legal, and regulatory inputs as applicable.
   - Trace each important design input to a user need, stakeholder expectation, risk, standard, or other credible source.
3. Concept-development process and preliminary analysis
   - Describe the structured process your team will use to generate, analyze, and compare multiple design concepts.
   - Use relevant theory, calculations, models, or simulations to investigate feasibility to an appropriate preliminary depth.
   - Consider uncertainty and trade-offs, including societal, environmental, and legal factors that could affect concept selection.
   - Do not present or justify the final selected concept in this report; final concept selection belongs in Progress Report 3.
4. **Preliminary risk analysis**
   - Identify early health and safety risks associated with implementation, prototyping, verification, manufacturing, deployment, or use.
   - Identify potentially applicable standards, codes, laws, and regulatory requirements, including the consequences of non-compliance. At this stage, a preliminary assessment is expected.
   - Identify relevant ethical concerns and their possible effects on users or society.
   - Comment on how the identified risks or constraints influence design concept generation and evaluation.

IBEHS 5P06: Progress Report 3 – Design Outputs and Configuration

Due Date: Friday, December 11, 2026
Length: 5–10 pages, excluding the title page, references, and appendices

Progress Report 3 documents the transition from design exploration to a justified final concept and preliminary configuration. It must define detailed design outputs, show the analytical basis for the selected concept, and establish a practical risk, compliance, and mitigation strategy for subsequent prototyping and verification.

Submission requirements

1. Summary

- Provide a concise overview of the problem, user need, selected solution direction, planned implementation, and expected impact.
- Include only the background from earlier reports that is necessary to understand this stage of the project.

2. Design outputs

- Present detailed, objective, and appropriately quantitative design outputs derived from the design inputs.
- Justify outputs using stakeholder evidence, literature, engineering analysis, standards, and disciplinary guidance.
- Address technical, economic, environmental, ergonomic, societal, safety, and regulatory considerations where relevant.

3. Final concept and preliminary design configuration

- Justify selection of the final design concept by tracing the decision to user needs, design inputs, evaluation criteria, analytical work, and documented trade-offs.
- Show a credible starting configuration using calculations, models, simulations, low-fidelity prototypes, schematics, pseudocode, or working demonstrations as appropriate.
- Incorporate the findings of your proof-of-concept to demonstrate the feasibility of your final design concept.
- Explain assumptions, limitations, constraints, feasibility, required resources, unanswered questions, and next steps.
- If an output will not be implemented, identify it explicitly and justify the omission using documented evidence.
- If the user need or project framing changed, explain the evidence and obtain instructor approval for the change.

4. Risk and compliance management plan

- Meet with your disciplinary instructor to identify and approve an appropriate risk management tool for your project. Use the approved tool to develop a risk management plan. If no specific tool is recommended, use a Failure Mode and Effects Analysis (FMEA)
- Identify health and safety risks associated with implementation, prototyping, verification, manufacturing, deployment, and use.
- Define feasible risk-mitigation and compliance strategies tied to applicable standards, codes, laws, and regulatory requirements.
- State which strategies must be implemented and verified in the final prototype.

IBEHS 5P06: Progress Report 4 – Prototype and Verification Plans

Due Date: Friday, February 5, 2027
Length: 4 – 5 pages, excluding the title page, references, and appendices

Progress Report 4 presents an executable plan for building the final prototype and verifying its performance. The report should make discipline-specific responsibilities and technical contributions visible, connect verification methods to the implemented design outputs, and incorporate the risk-mitigation and compliance strategies established in Progress Report 3.

Submission requirements

1. Summary

- Briefly restate the problem, user need, selected design, current configuration, implementation direction, and expected impact.
- Include only information from Progress Reports 1–3 needed to understand the prototype and verification plans.

2. Prototype plan

- Describe how the team will implement and build the final design, supported by schematics, images, architecture diagrams, workflows, or other appropriate representations.
- Identify critical components, interfaces, materials, tools, facilities, data, software, and other required resources.
- Show an equitable distribution of discipline-specific technical work and individual responsibilities. Confirm the expected technical scope with the appropriate disciplinary instructors.
- Explain how the compliance and mitigation strategies from Progress Report 3 will be incorporated.
- Identify assumptions, limitations, constraints, dependencies, likely setbacks, and contingency plans.

3. Verification plan

- Provide a verification method for every design output that will be implemented.
- Define the metric, acceptance criterion, procedure, required equipment or data, analysis method, and expected evidence for each verification activity.
- Use appropriate calculations, models, simulations, experiments, or other methods and identify how results will be corroborated through an independent method or source where feasible.
- Include plans to verify feasible risk-mitigation and compliance strategies.
- If design outputs have changed, present and justify the revisions. Outputs formally omitted in Progress Report 3 do not require verification.
- Execution of the full prototype and verification plan is not required at this submission stage.